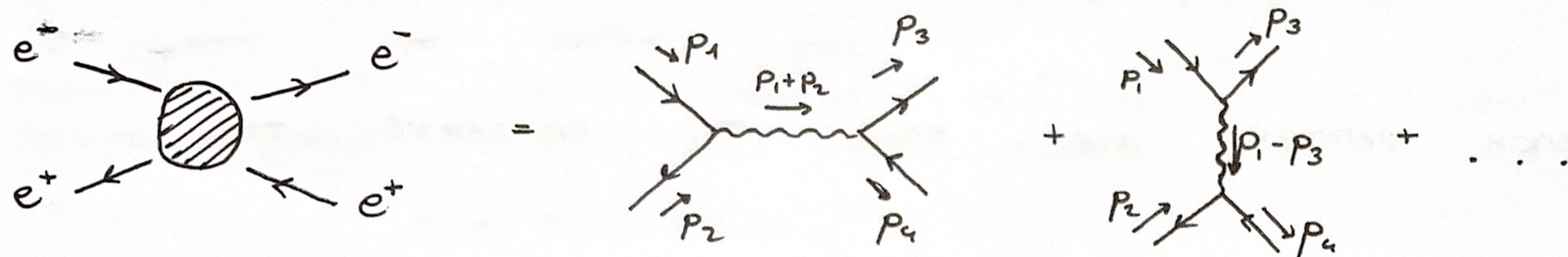


ADVANCED QUANTUM FIELD THEORY - OVERVIEW

WE ALREADY KNOW, FROM PREVIOUS COURSES, THAT WE CAN DO PERTURBATION THEORY OF A PARTICULAR PROCESS. FOR EXAMPLE, WE CAN COMPUTE:



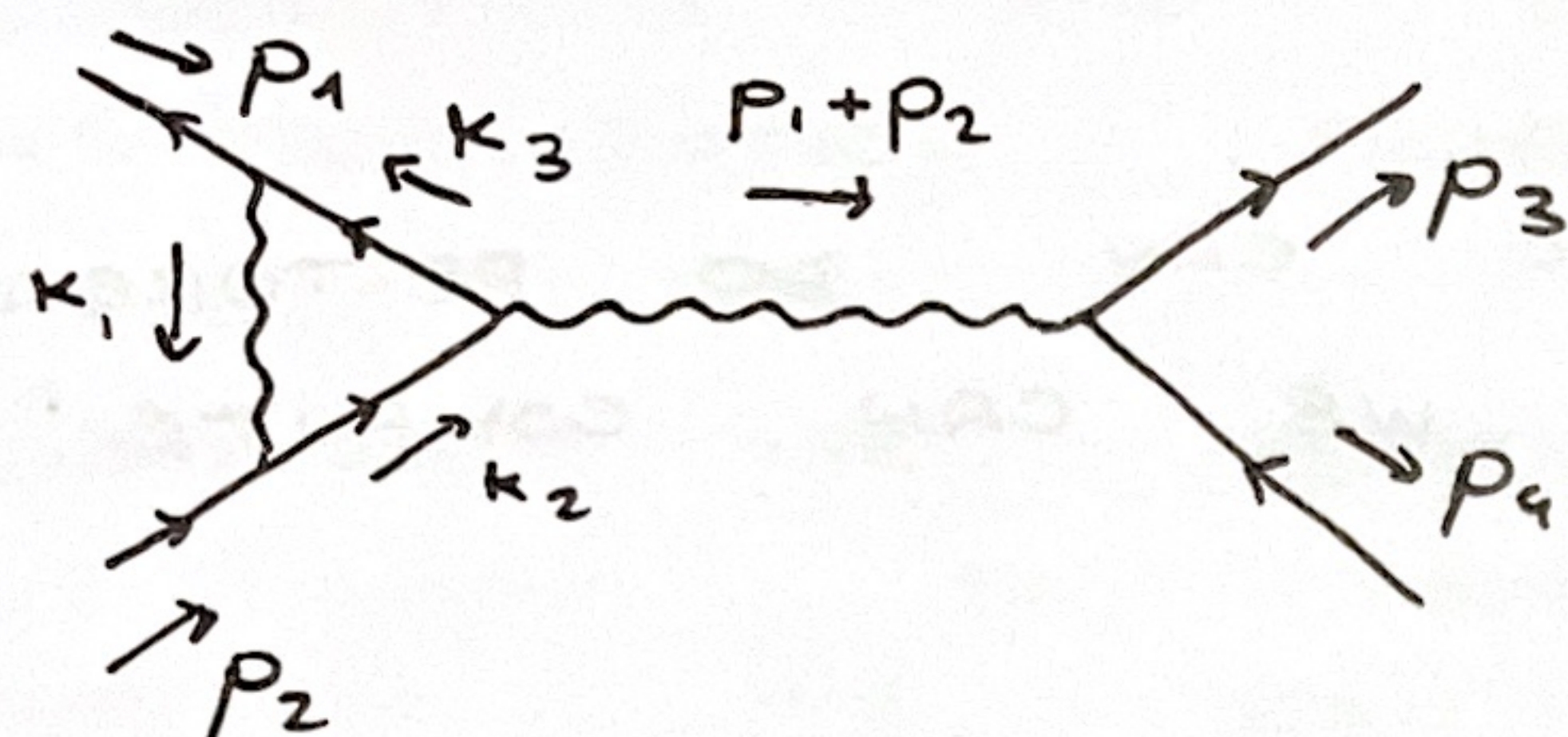
BHABHA SCATTERING

THE PROBLEM, THAT ARISE QUITE EASILY IN CALCULATION, IS THAT SOME OF THE DIAGRAMS DIVERGES. SOME OF THIS INFINITIES ARE NOTHING TO WORRY ABOUT BECAUSE THEY CANCEL OUT CONSIDERING OTHER RELATED PROCESSES, HOWEVER THERE ARE SOME INFINITIES LEFT OVER AND SEEM TO CAUSE MASSIVE PROBLEM. FOR EXAMPLE, IF WE COMPUTE THE SCATTERING CROSS SECTION FOR A PARTICULAR PROCESS, BUT IT CONTAINS ONE (OR MORE) FEYNMAN DIAGRAM THAT DIVERGES, IT'S NOT A GOOD THING HAVE AN INFINITY PROBABILITY TO OBSERVE A PROCESS.

WE CAN CURE THE THEORY TO HAVE A MEANINGFUL OBJECT IN OUR HANDS. WE NEED TO ANALYZE THE THEORY OF RENORMALIZATION IN PARTICULAR WE WILL STUDY THE PERTURBATIVE RENORMALIZATION.

THE RENORMALIZATION IS A LITTLE BIT TOUCHY AT FIRST SIGHT, SO IT HELPS A LOT SEE AN EXAMPLE FROM ANOTHER MATH ARGUMENT. BUT FIRST LET'S SEE FROM WHERE THIS DIVERGENSES COME FROM.

CONSIDER THE FEYNMAN DIAGRAM:



WHICH IS A SECOND ORDER CORRECTION TO BHABHA SCATTERING.

WE KNOW THAT IN EACH VERTEX WE HAVE TO CONSERVE THE MOMENTUM:

$$\begin{cases} p_1 + k_3 = k_1 \\ p_2 + k_1 = k_2 \\ k_2 = k_3 + p_3 + p_4 = k_3 + p_1 + p_2 \end{cases}$$

AND WE ALSO KNOW THAT NOT ALL THE MOMENTUM OF INTERNAL LINES ARE FIXED; IN FACTS FOR EACH LOOP (CLOSED) WE HAVE IN THE DIAGRAM WE GET AN UNSPECIFIED MOMENTUM, AND SINCE WE HAVE TO ADD UP ALL POSSIBILITIES FOR INTERNAL STATE, WE HAVE TO INTEGRATE OVER k .

THE INTEGRAL IS THE REASON WHY WE GET INFINITIES AND THEY CAN ARISE IN DIFFERENT REGION (OF k). IN PARTICULAR WE SAY:

⊕ IR DIVERGENCES: ARISE FROM REGION WHERE $k \rightarrow 0$.

⊕ UV DIVERGENCES: ARISE FROM REGION WHERE $k \rightarrow \infty$.

THIS DIVERGENCES DON'T ARISE IN EVERY DIAGRAM WITH A LOOP IN IT, BUT IT DEPENDS ON THE NUMBER (AND TYPE) OF PROPAGATOR (INTERNAL LINES). WE WILL TALK DURING THE COURSE ABOUT THE SUPERFICIAL DEGREE OF DIVERGENCE, WHICH HELP US TO UNDERSTAND IF WE HAVE A DIVERGENCE.



LET'S SEE AN EXAMPLE, KNOWN AS SINGULAR PERTURBATION THEORY, TO UNDERSTAND THE PROCEDURE WE WILL FOLLOW DURING THE RENORMALIZATION PROCESS.

CONSIDER THE EQUATION: $\epsilon x^2 - 2x + 1 = 0$ WITH $\epsilon \ll 1$
WE WANT TO SOLVE FOR x , BUT WITHOUT USE THE QUADRATIC FORMULA. WE CAN'T SOLVE IT EXACTLY, BUT WE CAN DO IT PERTURBATIVELY. ON ϵ . WE GUESS:

$$x = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots$$

WE CAN PUT $x(\epsilon)$ IN THE ORIGINAL EQUATION AND SEE:

$$\begin{aligned} \epsilon (x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots)^2 - 2(x_0 + \epsilon x_1 + \epsilon^2 x_2 + \dots) + 1 &= 0 \\ \Rightarrow (-2x_0 + 1) + \epsilon(x_0^2 - 2x_1) + \epsilon^2(2x_0x_1 - 2x_2) + \dots &= 0 \end{aligned}$$

SO, WE CAN SOLVE THIS EQUATION ORDER BY ORDER IN ϵ :

$$\Rightarrow x_0 = \frac{1}{2}, \quad x_1 = \frac{1}{8}, \quad x_2 = \frac{1}{16}, \quad \dots$$

BUT WE FIND JUST ONE SOLUTION AND WE KNOW THAT THE ROOT OF THE ORIGINAL EQUATION ARE TWO.

WHAT HAPPENED WAS THAT WE TAKE THE LIMIT FOR $\epsilon \rightarrow 0$ IN $\epsilon x^2 - 2x + 1 = 0$ SO WE GET A LINEAR EQUATION WITH ONLY ONE ROOT.

BUT THERE IS ANOTHER WAY TO GET AN EQUATION, QUADRATIC, THAT IN $\epsilon \rightarrow 0$ LIMIT GIVES US THE TWO ROOT.

WE HAVE TO REDEFINE THE VARIABLE $x = \delta(\epsilon) y$, WHERE $\delta(\epsilon)$ IS A PARAMETER THAT DEPENDS ON ϵ AND WHICH WE CAN FIX IN ORDER TO SOLVE THE PROBLEMATIC BEHAVIOR. WE WILL SOLVE THE EQUATION FOR y INSTEAD OF x .

NOTE THAT $\delta(\epsilon)$ IS COMPLETELY ARBITRARY. WE HAVE THE EQUATION

$$\epsilon \delta(\epsilon)^2 y^2 - 2 \delta(\epsilon) y + 1 = 0$$

AND A GOOD CHOICE WE CAN MAKE IS $\delta(\epsilon) = \frac{1}{\epsilon}$ AND THEN MULTIPLY ALL BY ϵ :

$$y^2 - 2y + \epsilon = 0$$

WHICH IS PERFECTLY WELL BEHAVED FOR $\epsilon \rightarrow 0$. NOW, WE CAN DO THE SAME THING AS BEFORE, SO WRITE FOR y A PERTURBATIVE SOLUTION IN ϵ :

$$y = y_0 + \epsilon y_1 + \epsilon^2 y_2 + \dots$$

AND SOLVE ORDER BY ORDER:

$$\Rightarrow (y_0^2 - 2y_0) + \epsilon (2y_0 y_1 - 2y_1 + 1) + \epsilon^2 (y_1^2 + 2y_0 y_2 - 2y_2) + \dots = 0$$

$$\Rightarrow y^{(1)} = \frac{\epsilon}{2} + \frac{\epsilon^2}{8} + \dots, \quad y^{(2)} = 2 - \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + \dots$$

SO, IN THIS WAY, WE RECOVER OUR SECOND ROOT AND TO GET THE SOLUTION IN x WE JUST MULTIPLY THE ROOTS BY $\delta(\epsilon)$:

$$x^{(1)} = \frac{1}{2} + \frac{\epsilon}{8} + \dots, \quad x^{(2)} = \frac{2}{\epsilon} - \frac{1}{2} - \frac{\epsilon}{8} + \dots$$

FOR THE RENORMALIZATION OF QFT WE WILL FOLLOW A SIMILAR SCHEME AS THE PREVIOUS EXAMPLE. THE PROCEDURE WAS:

- 1 RE-DEFINE VARIABLE IN TERMS OF ARBITRARY SCALE PARAMETER.
- 2 FIX BY HAND BEHAVIOR OF THE PARAMETER IN ORDER TO ELIMINATE ARTIFICIALLY THE PROBLEMATIC BEHAVIOR.
- 3 SOLVE WELL BEHAVED PROBLEM AND FACTOR BACK IN THE SCALE PARAMETERS TO GET OUR TRUE SOLUTION.

ONE MORE THING. THE INTRODUCTION OF $\delta(\epsilon)$ WAS NECESSARY TO SOLVE THE EQUATION, BUT THE CHOICE $\delta(\epsilon) = 1/\epsilon$ WAS COMPLETELY ARBITRARY AND WE COULD DEFINE IT ADDING A FINITE CONSTANT a .

$$\Rightarrow \delta = \delta(\epsilon, a) \quad \Rightarrow \gamma = \gamma(\epsilon, a)$$

BUT a DOESN'T APPEAR IN OUR ORIGINAL PROBLEM, SO WHEN WE GO BACK TO X , IT CANNOT DEPEND ON a .

WE CAN WRITE A DIFFERENTIAL EQUATION ABOUT IT:

$$\frac{dx}{da} = \frac{d(\delta\gamma)}{da} = \frac{d\delta}{da} \gamma + \delta \frac{d\gamma}{da} \equiv 0$$

$$\Rightarrow \boxed{\delta \frac{d\gamma}{da} = - \frac{d\delta}{da} \gamma}$$

WHICH HAVE TO BE SATISFIED AT ANY ORDER IN PERTURBATION THEORY.

Now, FINALLY TALK ABOUT PARTICLE PHYSICS.

WE KNOW THAT IN OUR THEORY WE HAVE SOME PARAMETER NOT FIXED BY THE THEORY ITSELF, LIKE COUPLINGS ($g \dots$), MASSES (m) AND FIELD NORMALIZATIONS. WE USUALLY CALL THESE UN-FIXED PARAMETERS "BARE PARAMETERS".

DURING THE RENORMALIZATION PROCESS WE RESCALE THESE PARAMETERS:

$$g_0 = Z_g g_1$$

$$m_0 = Z_m m_1$$

$$\phi_0 = Z_\phi^{1/2} \phi_1$$

AND CALL g_0, m_0, ϕ_0 THE BARE PARAMETERS, Z_g, Z_m, Z_ϕ THE "RENORMALIZATION CONSTANTS" AND g_1, m_1, ϕ_1 THE "DRESSED PARAMETERS".

TYPICALLY WE WRITE:

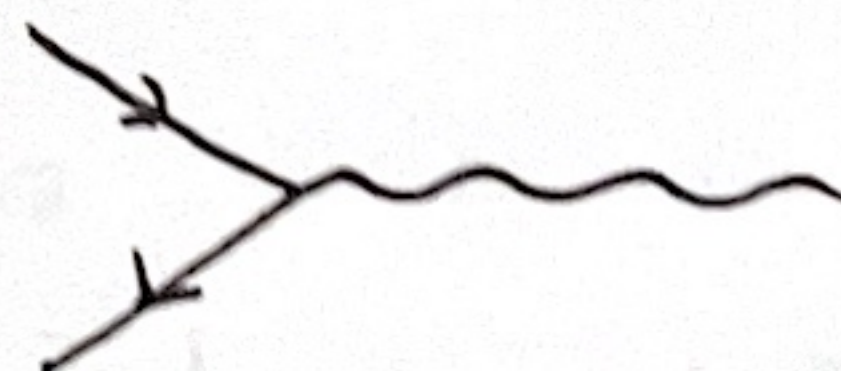
$$Z_g = 1 + \delta Z_g$$

$$Z_m = 1 + \delta Z_m$$

$$Z_\phi = 1 + \delta Z_\phi$$

WHERE WE CALL δZ "COUNTER-TERMS".

WITH THIS PROCEDURE WE CAN KEEP THE FEYNMAN RULES, BUT WITH BARE PARAMETERS REPLACED WITH THE RENORMALIZED COUNTERPARTS:



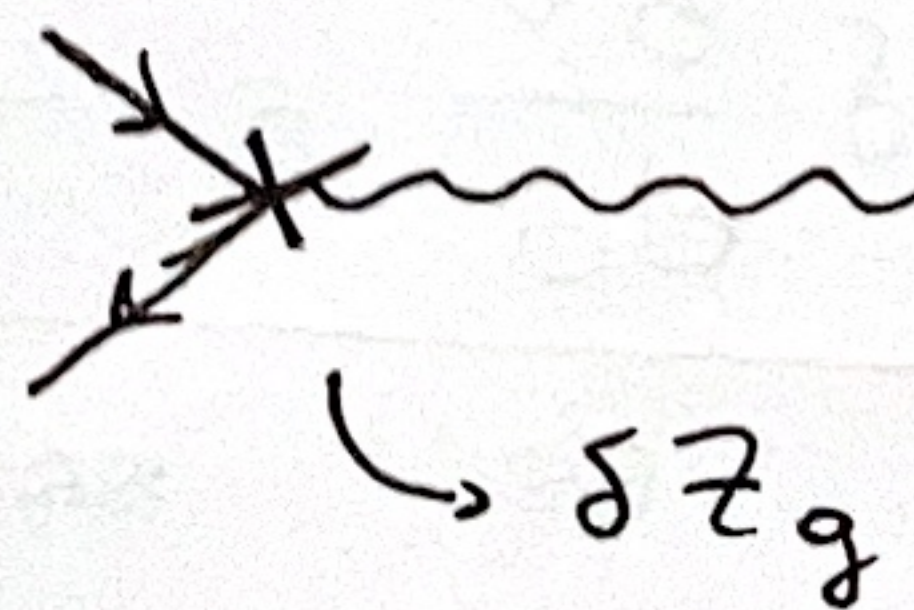
HOWEVER WE HAVE TO ADD NEW FEYNMAN RULES, WE DIDN'T HAVE BEFORE, RELATED TO COUNTER TERM:



$\delta Z_4, \delta Z_{m4}$



δZ_A



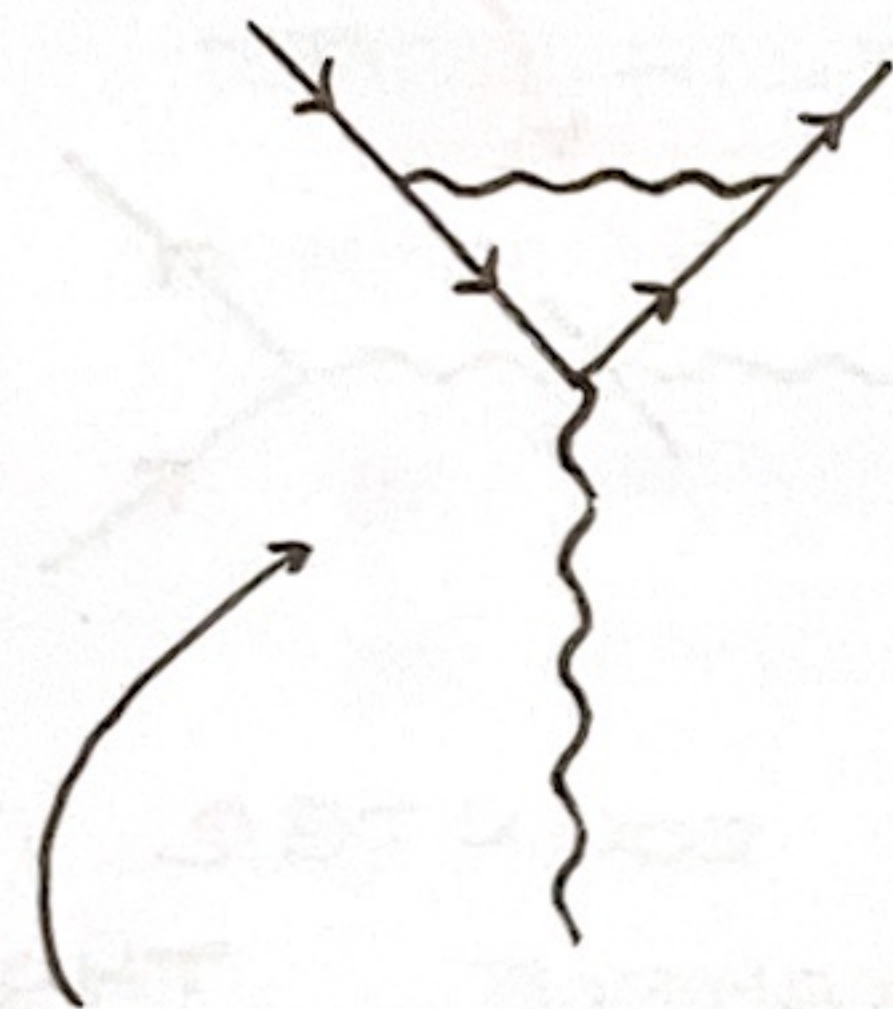
δZ_g

REMEMBER WE HAVE THE SAME THEORY AS BEFORE, BUT WE'VE JUST SHUFFLED THINGS AROUND TO MAKE IT LOOK SLIGHTLY DIFFERENT (AND SOLVE SOME PROBLEMS).

NOTE THE COUNTER TERMS HAVE THE SAME ROLE OF $\delta(\epsilon)$ IN THE EXAMPLE.

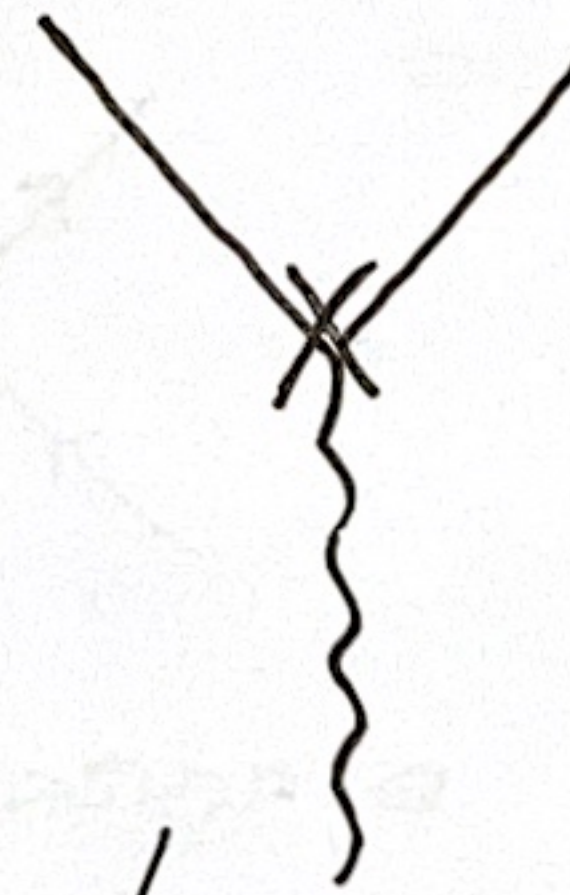
NOW, WE HAVE TO FIX THESE COUNTER TERM TO ELIMINATE UV DIVERGENCES (JUST LIKE WE FIXED δ TO ELIMINATE SINGULAR BEHAVIOR IN y^2 TERM).

WHEN WE CONSIDER A PROCESS, WITH SPECIFIC EXTERNAL LINES, WE CAN BUILD THE SIMILAR FOR THE COUNTER TERMS. FOR EXAMPLE:



LOOP CORRECTION TO QED VERTEX WHICH IS DIVERGENT FOR THE SAME REASON AS BEFORE.

+



COUNTER TERM DIAGRAM WHICH HAVE THE SAME INITIAL AND FINAL STATE OF THE "NORMAL" DIAGRAM.

AND IF WE REQUIRE THAT THIS SUM IS FINITE

$\Rightarrow$ WE FIX δZ_q (TO ONE-LOOP ORDER)

$$\delta Z = (\text{UV DIVERGENCE}) + (\text{FINITE})$$

GIVES US THE RENORMALIZATION SCHEME (MS, $\overline{\text{MS}}$ OR ON-SHELL SCHEME)

THEN WE CAN DO THE SAME THING WITH THE LOOP CORRECTION TO THE PROPAGATOR:



+



= FINITE

$\rightarrow$ FIXES

$\delta Z_4, \delta Z_{m_4}$



+



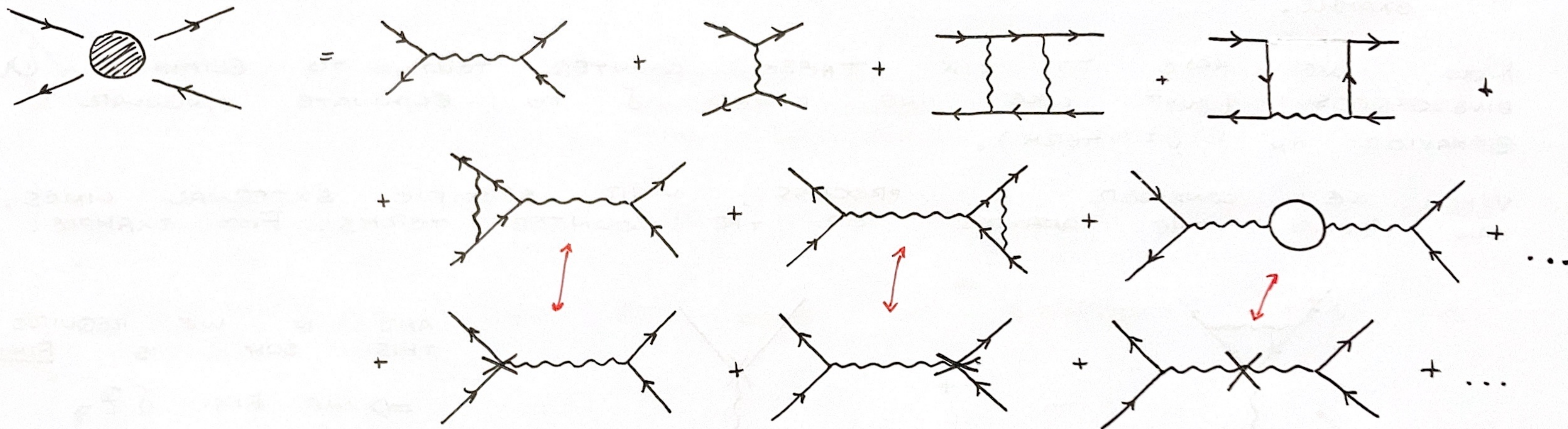
= FINITE

$\rightarrow$ FIXES

δZ_A

So, AT THIS POINT WE HAVE FIXED ALL THE COUNTER-TERM
 SO THAT THEY FIX ALL OUR DIVERGENCES IN OUR THEORY.

COMING BACK TO THE BHABHA SCATTERING WE CAN DO THE CALCULATION AS BEFORE:
 BUT USING THE DRESSED PARAMETERS!



WHERE WE INCLUDED ALL THE DIAGRAMS WITH THE COUNTER TERM
 (UP TO A GIVEN ORDER). IT'S EASY IN THIS EXAMPLE THAT WE
 CAN PAIR THE DIVERGENT DIAGRAM WITH THE CORRESPONDENT COUNTER TERM,
 AND THE SUM GIVES US (BECAUSE WE IMPOSE THAT) A FINITE
 QUANTITY. SO, AT THE END WE GET A FINITE AMPLITUDE FOR
 THE PROCESS.

SO FAR WHAT WE DID IS:

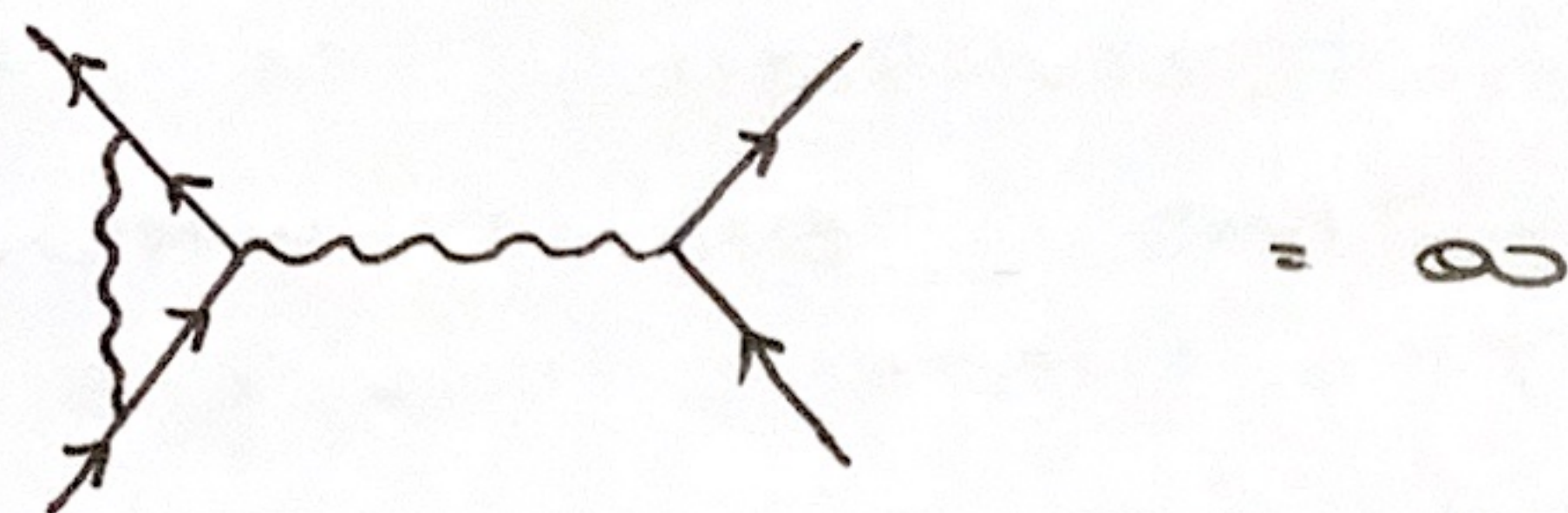
1. RE-SCALE UNSPECIFIED PARAMETERS IN THEORY ($g_0 = (1 + \delta Z_g) g_1 \dots$)

2. FIX COUNTER-TERMS TO CANCEL SPECIFIC UV DIVERGENCES IN THE THEORY

3. "MULTIPLY" BY SCALE FACTORS BY INCLUDING FEYNMAN DIAGRAMS WHICH INCLUDE COUNTERTERM INSERTIONS.

WITH THIS PROCEDURE WE CAN MAKE THE SCATTERING AMPLITUDE FINITE.

UNFORTUNATELY THIS IS NOT THE END OF THE STORY (WE DID A COURSE FOR THAT). WHEN WE HAVE A DIVERGENT DIAGRAM, SUCH AS:



WE STILL HAVE IMPORTANT FINITE BEHAVIORS IN IT WE NEED TO CONSIDER (ALSO THE STRUCTURE OF THE DIVERGENCE IS IMPORTANT).

SO WE HAVE TO REGULATE THE DIVERGENCE SOME HOW TO FORCE TO GIVE US A FINITE RESULT.

THIS REGULARIZATION CAN BE DONE IN DIFFERENT WAYS, THE MOST POPULAR ARE:

⊕ MOMENTUM CUT-OFF Λ

⊕ DIMENSIONAL REGULARIZATION $\rightarrow$ WE GO IN ARBITRARY DIMENSION $d = 4 - 2\epsilon$ AND TAKE $\epsilon \rightarrow 0$ AT THE END OF CALCULATION.

WITH THESE METHODS WE INTRODUCE AN ENERGY SCALE, WHICH IS ARTIFICIAL.

SO, TO REGULATE INTEGRALS, AND FIND THE COUNTER TERMS, WE HAVE TO (SOMEHOW) INTRODUCE A NEW ARTIFICIAL ENERGY SCALE μ , WHICH, SINCE IT DOESN'T APPEAR IN THE ORIGINAL THEORY, THE BARE PARAMETERS CANNOT DEPEND ON μ (FOR THE SAME REASON χ COULDN'T DEPEND ON d IN THE EXAMPLE). AS BEFORE WE GET A SET OF DIFFERENTIAL EQUATIONS:

$$\frac{dg}{d \log \mu} = -g \frac{1}{Z_g} \frac{dZ_g}{d \log \mu}, \quad \frac{dm}{d \log \mu} = -m \frac{1}{Z_m} \frac{dZ_m}{d \log \mu}$$

THIS SET OF DIFFERENTIAL EQUATIONS ARE CALLED "RENORMALIZATION GROUP" EQUATIONS. THESE EQUATIONS GUARANTEE THAT PHYSICAL QUANTITIES ARE COMPLETELY INDEPENDENT FROM THIS ARTIFICIAL ENERGY SCALE.

NOW WE HAVE THAT THE RENORMALIZED PARAMETERS, IN GENERAL, DEPEND ON ENERGY SCALE μ , AND WHEN WE DO PERTURBATION SERIES, NOW IN TERMS OF g (NOT g_0), CAN HAPPEN THAT EVEN IF g IS SMALL AT SOME SCALE μ_0 , AT ANOTHER SCALE μ_1 , g MAY BECOME LARGE AND THE PERTURBATION THEORY BREAK DOWN IN THIS REGION. THIS IS WHAT HAPPENS FOR LOW-ENERGY QCD.

IN ADDITION, TO FIX THE UV DIVERGENCE OF OUR THEORY WE CAN WORK WITH THE COUNTER TERMS, AS WE SAW. BUT IT'S POSSIBLE THAT SOME INFINITIES REMAIN IN THE THEORY AND TO FIX THAT WE HAVE TO INCLUDE NEW INTERACTION (NEW VERTICES). THIS HASN'T TO BOTHER US BECAUSE WHAT WE ARE SAYING ADDING NEW INTERACTION IS THAT IN THE ENERGY REGION, WHERE WE HAVE THE INFINITY, THERE ARE SOME PROCESS THAT ARISE ONLY THERE.

BUT WE CAN HAVE RENORMALIZABLE THEORY, WHERE WE NEED TO ADD A FINITE NUMBER OF NEW INTERACTION, AND NON-RENORMALIZABLE THEORY WHERE WE NEED TO ADD INFINITE NEW INTERACTION.

NOTE ADDING NEW INTERACTION WE CAN BUILD NEW COUNTER-TERM THAT WE CAN USE TO REGULATE THE THEORY.

THE NON-RENORMALIZABLE THEORY ARE NOT USELESS, BUT WE CAN DEFINE THEM AS EFFECTIVE FIELD THEORY, SO AS THEORY VALID UP TO A SCALE Λ (WHERE THE NUMBER OF COUNTER-TERMS WE NEED TO INCLUDE ARE FINITE).

→ FOR EXAMPLE THE FERMI THEORY FOR ELECTROWEAK INTERACTIONS.

(X)